

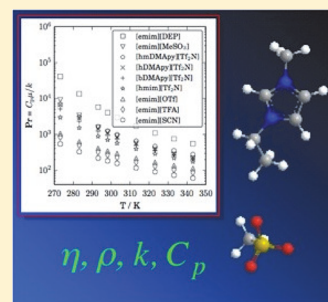
A Computational and Experimental Study of the Heat Transfer Properties of Nine Different Ionic Liquids

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S Supporting Information

ABSTRACT: New experimental thermal conductivity, density, viscosity, glass transition temperature, and heat capacity values were measured for nine ionic liquids (ILs): [emim][TFA], [emim][OTf], [emim][DEP], [emim][MeSO₃], [emim][SCN], [hmim][Tf₂N], [bDMApy][Tf₂N], [hDMApy][Tf₂N], and [hmDMApy][Tf₂N]. Classical molecular mechanics force fields were developed and used to calculate thermodynamic and transport properties for these ILs using molecular dynamics. Two versions of each force field were developed: one with integer charges of ± 1 and one with all charges scaled by 0.8. The force fields with total charges of ± 0.8 generally gave better agreement with experimental results. Very good agreement was obtained for density and heat capacity. Simulated values for thermal conductivity slightly overpredicted experimental results but captured trends between different ILs very well. Experimental Prandtl numbers were determined as a function of temperature and can exceed 10 000 at low temperature. Prandtl numbers on the order of 100–1000 were observed above 330 K. These values suggest that heat transfer with ionic liquids will be dominated by convective effects.



INTRODUCTION

Early on, Wilkes¹ discussed the possibility of using ionic liquids (ILs) as thermal fluids, particularly for high temperature heat transfer applications such as solar thermal energy generation. Stimulated mainly by their low volatility and high thermal stability, several other authors have also raised the possibility of using ILs for this and related purposes.^{2–5} Some of the key thermophysical properties necessary for assessing the feasibility of using ILs as thermal fluids have been measured. These properties include thermal conductivity, density, viscosity, melting point/glass transition temperature, and heat capacity. In general, experimental densities and viscosities tend to be more available than the other properties, and often there are missing data for particular ILs or the properties are not measured under conditions of interest. Depending on the application, one or more of these properties will be critically important, while others may be less so.

Being able to understand and even predict why a given IL has a particular set of properties is important, because the vast number of potential ILs makes it impossible to make and characterize all of them. Ideally, one would like to perform a limited number of experiments on a carefully selected set of ILs and use this information to try to understand structure–property relationships. This is the objective of the present work. Nine different ILs were obtained and new experimental thermal conductivity, density, viscosity, glass transition temperature, and heat capacity data were measured.

Table 1 summarizes the chemical structures, full names, and abbreviations of the cations and anions investigated in this study. Using their abbreviated names, the nine ILs investigated are [emim][TFA], [emim][OTf], [emim][DEP], [emim][MeSO₃],

[emim][SCN], [hmim][Tf₂N], [bDMApy][Tf₂N], [hDMApy][Tf₂N], and [hmDMApy][Tf₂N].

To gain even more insight, these experiments were complemented with atomistic-level simulations of the same ILs. For each cation and anion listed above, a classical force field was developed and molecular dynamics simulations were performed to calculate all of the properties measured experimentally except the melting point/glass transition temperature. By comparing the simulation results with the experimental data, it is possible to assess how accurate such simulations can be in predicting these properties. If the simulations are shown to be accurate, they are likely to be accurate in predicting properties for other combinations of the cations and anions not studied experimentally, as well as at conditions difficult to access experimentally, such as very high temperatures. Moreover, the detailed atomistic information available from the simulations can help provide deeper understanding as to why certain cations and anions impact these properties as they do.

EXPERIMENTAL DETAILS

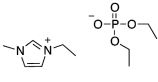
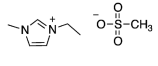
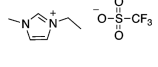
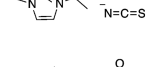
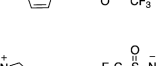
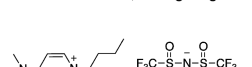
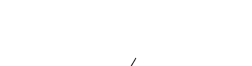
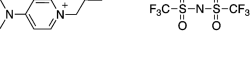
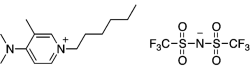
Synthesis. All the ILs except [hmDMApy][Tf₂N] and [hmim][Tf₂N] were purchased from commercial sources and used as received. [bDMApy][Tf₂N] (> 0.98 weight fraction purity), [hDMApy][Tf₂N] (> 0.98 weight fraction) and [emim][TFA] (> 0.99 weight fraction) were purchased from Merck KGaA, [emim][OTf] (> 0.99 weight fraction) from Solvent Innovation, [emim][SCN] (> 0.99 weight fraction) from

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Table 1. Ionic Liquids Used in This Study

chemical structure	full name	abbreviation	weight fraction
	1-ethyl-3-methylimidazolium diethylphosphate	[emim][DEP]	0.99 ^a 0.98 ^b
	1-ethyl-3-methylimidazolium methanesulfonate	[emim][MeSO ₃]	0.99
	1-ethyl-3-methylimidazolium trifluoromethanesulfonate	[emim][OTf]	0.99
	1-ethyl-3-methylimidazolium thiocyanate	[emim][SCN]	0.99
	1-ethyl-3-methylimidazolium trifluoroacetate	[emim][TFA]	0.99
	1-hexyl-3-methylimidazolium bis(trifluoromethylsulfonyl)imide	[hmim][Tf ₂ N]	0.99
	1-butyl-4-(dimethylamino)pyridinium bis(trifluoromethylsulfonyl)imide	[bDMApy][Tf ₂ N]	0.98
	1-hexyl-4-(dimethylamino)pyridinium bis(trifluoromethylsulfonyl)imide	[hDMApy][Tf ₂ N]	0.98
	1-hexyl-3-methyl-4-(dimethylamino)pyridinium bis(trifluoromethylsulfonyl)imide	[hmDMApy][Tf ₂ N]	0.99

^aUsed for the viscosity measurements. ^bUsed for the heat capacity measurements.

Fluka and [emim][MeSO₃] (> 0.99 weight fraction) from Iolitec. The [emim][DEP] used for the heat capacity measurement was purchased from Iolitec (> 0.98 weight fraction), while the sample used for the viscosity measurement was from Merck KGaA (> 0.99 weight fraction). [hmDMApy][Tf₂N] and [hmim][Tf₂N] (both > 0.99 weight fraction) were synthesized in our laboratory using previously described procedures.^{6,7} The samples were dried under reduced pressure (10 mbar) for at least 48 h before each measurement and the water content was determined by a Metrohm 831 coulometric Karl Fischer titrator. The measured water contents of samples used for various experimental measurements are listed in the Supporting Information.

Thermal Conductivity. Thermal conductivity measurements were made on an Anter Corporation UNITHERM 2022 Guarded Heat Flow Meter and Thermal Conductivity Measuring System. This system utilizes a steady-state measurement technique in which heat flows from the upper surface of the sample to the lower surface, establishing an axial temperature gradient. Once thermal equilibrium is achieved, the heat flow and temperature difference across the sample, along with the sample's thickness, are used to determine thermal conductivity. The relative standard uncertainty is $u_r(k) = 0.08$.

The UNITHERM 2022 system was calibrated for use with liquids over the test range of 293 K to 373 K. The calibration was validated by measuring the thermal conductivity of [hmim][Tf₂N] and comparing it to values already reported in the literature by Ge et al.⁸ The experimental results were within 0.5 % of those reported by Ge at the calibration temperature.

Density. Density measurements were performed at atmospheric pressure in a DMA 4500 Anton Paar oscillating U-tube

Table 2. Experimental Thermal Conductivity k (Mean Value, 300 K to 375 K) at Pressure $p = 0.1$ MPa^a

ionic liquid	$k/W \cdot m^{-1} \cdot K^{-1}$
[emim][DEP]	0.19
[emim][MeSO ₃]	0.21
[emim][OTf]	0.16
[emim][SCN]	0.21
[emim][TFA]	0.17
[hmim][Tf ₂ N]	0.12
[bDMApy][Tf ₂ N]	0.13
[hDMApy][Tf ₂ N]	0.13
[hmDMApy][Tf ₂ N]	0.12

^aRelative standard uncertainty is $u_r(k) = 0.08$.

densitometer, which includes an automatic correction for the viscosity of the sample, as described previously.⁹ The temperature was controlled using two integrated Pt 100 platinum resistance thermometers. The relative standard uncertainties are $u_r(\rho) = 0.0015$ for [hmim][Tf₂N] and [hmDMApy][Tf₂N], and $u_r(\rho) = 0.003$ for [bDMApy][Tf₂N] and [hDMApy][Tf₂N].

Viscosity. Viscosity measurements were carried out with a Brookfield DV-III Ultra (cone and plate) rheometer, as described previously.⁹ The temperature was controlled using a TC-602 bath thermostat with a Brookfield temperature controller unit attached. During the sample measurement the rheometer chamber was under an inert nitrogen atmosphere. However, Karl Fisher titration showed that after the viscosity measurements there was a rise in sample water content of $2 \cdot 10^{-4}$ weight

fraction for the least hydrophilic ILs and $3 \cdot 10^{-3}$ weight fraction for the most hydrophilic ILs. The relative standard uncertainty for this measurement is $u_r(\eta) = 0.05$.

Heat Capacity and Glass Transition Temperature. Glass transition temperature T_g and heat capacity C_p values were measured using a Mettler-Toledo differential scanning calorimeter (DSC), model DSC822e with STARE software data collection, using a procedure described previously.⁶ We report a standard uncertainty of $u(T_g) = 3$ K, which takes into account the inherent instrument uncertainty and the impurities in the ILs. The heat capacities were determined by the sapphire method. The relative standard uncertainty for this measurement is $u_r(C_p) = 0.05$.

Simulation Details. A simple “class I” force field model was used to describe bonded and nonbonded interactions between atoms

$$E_{\text{total}} = \sum_{\text{bonds}} K_r(r - r_0)^2 + \sum_{\text{angles}} K_\theta(\theta - \theta_0)^2 + \sum_{\text{dihedrals}} K_{\phi,n}[1 + \cos(n\phi - \gamma)] + \sum_{i < j} 4\epsilon_{ij} \left[\left(\frac{\sigma_{ij}}{r_{ij}} \right)^{12} - \left(\frac{\sigma_{ij}}{r_{ij}} \right)^6 \right] + \sum_{i < j} \frac{q_i q_j}{r_{ij}} \quad (1)$$

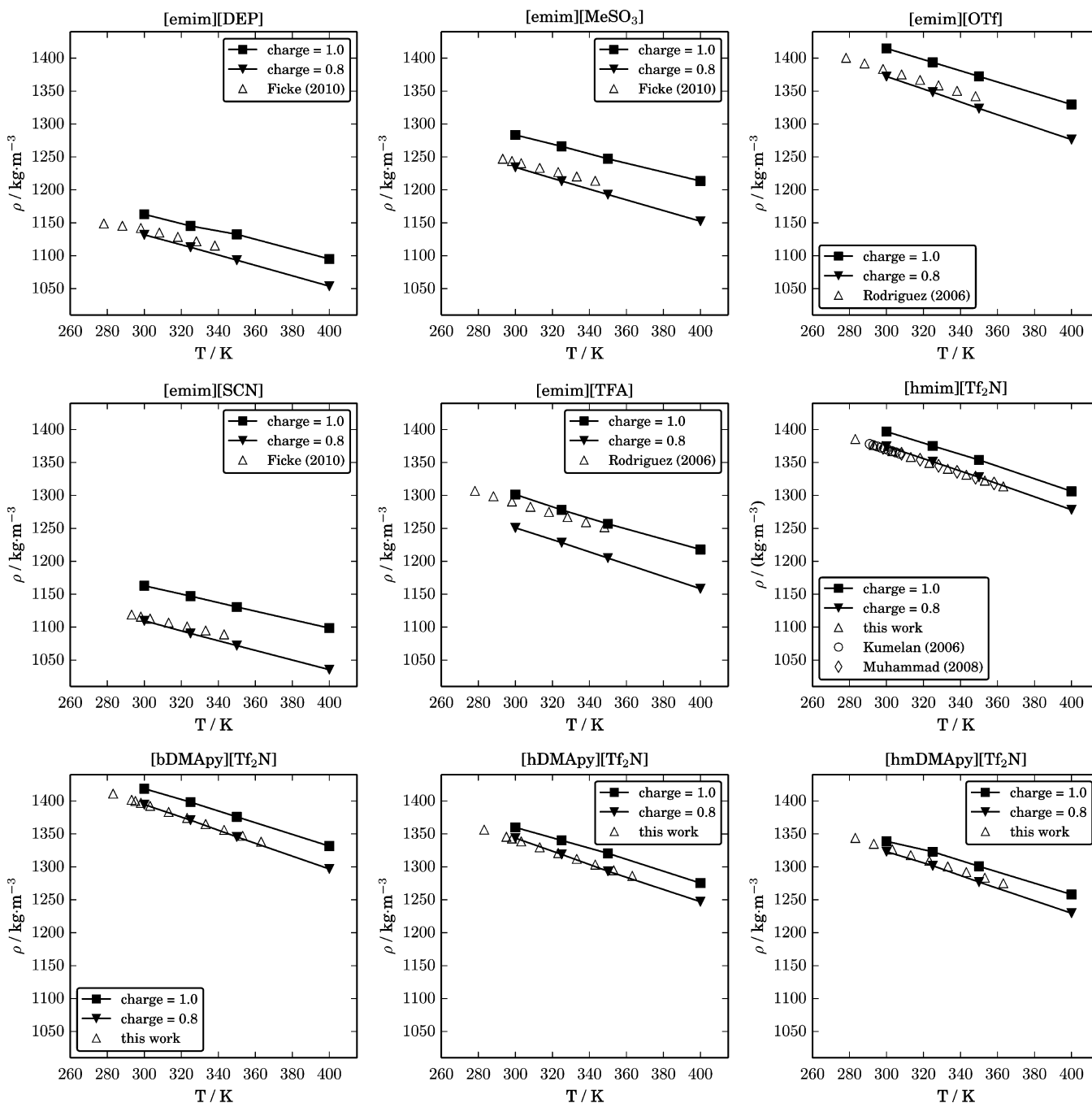


Figure 1. Density versus temperature from NPT molecular dynamics simulations (filled symbols) using force fields with unscaled and 0.8-scaled charges compared to experimental results (open symbols).^{9,24–26} Simulation standard uncertainty $u(\rho)$ is $10 \text{ kg} \cdot \text{m}^{-3}$. Lines through simulation results are guides to the eye.

Table 3. Experimental Density ρ versus Temperature at Pressure $p = 0.1 \text{ MPa}^a$

[hmim][Tf ₂ N]		[bDMApy][Tf ₂ N]		[hDMApy][Tf ₂ N]		[hmDMApy][Tf ₂ N]	
T/K	$\rho/\text{kg}\cdot\text{m}^{-3}$	T/K	$\rho/\text{kg}\cdot\text{m}^{-3}$	T/K	$\rho/\text{kg}\cdot\text{m}^{-3}$	T/K	$\rho/\text{kg}\cdot\text{m}^{-3}$
283	1385.63	283	1411.09	283	1356.57	283	1342.45
293	1376.50	293	1401.77	295	1345.66	293	1333.67
303	1367.38	295	1399.92	298	1342.99	303	1325.08
313	1358.30	298	1397.13	303	1338.54	313	1316.49
323	1349.27	303	1392.52	313	1329.67	323	1307.95
333	1340.34	313	1383.33	323	1320.85	333	1299.45
343	1331.41	323	1374.17	333	1312.10	343	1290.98
353	1322.54	333	1365.07	343	1303.40	353	1282.55
363	1313.73	343	1356.06	353	1294.76	363	1274.19
		353	1347.10	363	1286.19		
		363	1338.20				

^aRelative standard uncertainties are $u_r(\rho) = 0.0015$ for [hmim][Tf₂N] and [hmDMApy][Tf₂N], and $u_r(\rho) = 0.003$ for [bDMApy][Tf₂N] and [hDMApy][Tf₂N].

Table 4. Experimental Viscosity η versus Temperature at Pressure $p = 0.1 \text{ MPa}^a$

T/K	[emim][DEP]	[emim][MeSO ₃]	[bDMApy][Tf ₂ N]
	$\eta/\text{mPa}\cdot\text{s}$	$\eta/\text{mPa}\cdot\text{s}$	$\eta/\text{mPa}\cdot\text{s}$
283	1280	450	198
293	580	210	108
298	410	155	83
303	300	117	65
313	168	70	42
323	101	45	29
333	66	31	21
343	45	23	15.7

^aRelative standard uncertainty is $u_r(\eta) = 0.05$.

Table 5. Experimental Glass Transition Temperature T_g at Pressure $p = 0.1 \text{ MPa}^a$

ionic liquid	T_g/K
[emim][DEP]	207
[emim][MeSO ₃]	211
[bDMApy][Tf ₂ N]	205

^aStandard Uncertainty is $u(T_g) = 3 \text{ K}$.

The bonded interaction parameters K_v , K_θ , and $K_{\phi,n}$ are force constants, r_0 and θ_0 are equilibrium structure parameters, and n and γ are periodicity and phase angle, respectively, for dihedral interactions. Note that the summation over dihedrals includes both conventional and “improper” dihedrals. Accounting for nonbonded contributions to the potential energy, the Lennard-Jones parameters ϵ and σ describe repulsive and dispersive interactions, and the point charges q describe electrostatic interactions. For nonbonded interactions between different atom types, the parameters ϵ_{ij} and σ_{ij} were calculated using Lorentz–Berthelot combining rules

$$\epsilon_{ij} = \sqrt{\epsilon_{ii}\epsilon_{jj}}$$

$$\sigma_{ij} = \frac{\sigma_{ii} + \sigma_{jj}}{2} \quad (2)$$

All force field parameter values for the molecules simulated in this work, along with details on how the force fields were parametrized, are provided in the Supporting Information. In this study two separate force fields were developed for each IL. The first force field used partial atomic charges under the assumption

Table 6. Vogel–Fulcher–Tamman Model Parameters (eq 3) Resulting from Fit to Viscosity Data in 4

ionic liquid	$\eta_0/\text{mPa}\cdot\text{s}$	k/K
[emim][DEP]	0.783	563
[emim][MeSO ₃]	0.843	453
[bDMApy][Tf ₂ N]	0.779	432

Table 7. Experimental Heat Capacity C_p versus Temperature at Pressure $p = 0.1 \text{ MPa}^a$

T/K	[emim][DEP]	[bDMApy][Tf ₂ N]	[hDMApy][Tf ₂ N]
	$C_p/\text{J}\cdot\text{K}^{-1}\cdot\text{kg}^{-1}$	$C_p/\text{J}\cdot\text{K}^{-1}\cdot\text{kg}^{-1}$	$C_p/\text{J}\cdot\text{K}^{-1}\cdot\text{kg}^{-1}$
298	1740	1330	1360
303	1760	1340	1370
308	1770	1350	1380
313	1780	1350	1390
318	1800	1360	1400
323	1810	1370	1410
328	1820	1380	1430
333	1840	1390	1440
338	1850	1400	1450
343	1870	1410	1460

^aRelative standard uncertainty is $u_r(C_p) = 0.05$.

that the cations and anions had a formal charge of ± 1 . The second force field was exactly the same as the first except every partial charge was scaled by a factor of 0.8, which is equivalent to reducing the potential energy of every electrostatic interaction by a factor of 0.8^2 or 0.64. The reason for doing this is that in previous studies, accounting explicitly for ion polarizability was generally found to result in much better agreement between experimental and simulated dynamical properties such as diffusivity and viscosity.^{10–14} Scaling partial charges on the ions has been shown to be effective at capturing this effect as well as charge transfer effects thought to occur between cations and anions in the condensed phase^{15–20} but with no additional computational cost.

All simulations for this study were performed using the open source LAMMPS molecular dynamics (MD) software package.^{21,22} Details of how the simulations were carried out, including example LAMMPS input scripts demonstrating specific simulation settings, are provided in the Supporting Information.

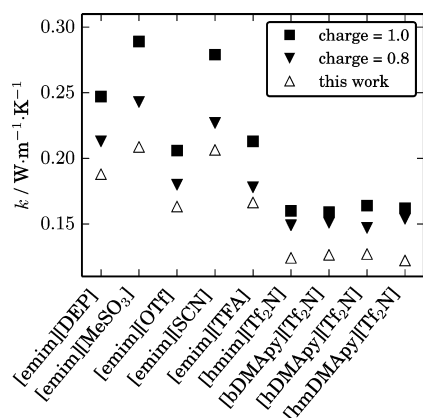


Figure 2. Thermal conductivity from RNEMD simulations (at 400 K, filled symbols) using force fields with unscaled and 0.8-scaled charges compared to experimental results (mean from 300 to 375 K, open symbols). Error bars are omitted for clarity. Simulation standard uncertainty $u(k)$ is $0.015 \text{ W} \cdot \text{m}^{-1} \cdot \text{K}^{-1}$.

EXPERIMENTAL RESULTS

Table 2 summarizes experimentally measured thermal conductivities for each IL in this study. Thermal conductivity was measured at several temperatures between 300 K and 375 K, but the temperature dependence was marginal as has been observed previously with ionic liquids.¹ Within the uncertainty of the experimental procedure, all results for a given IL were equivalent, and the mean value of those results is reported. ILs containing the $[\text{Tf}_2\text{N}]^-$ anion and $[\text{hmim}]^+$ or $[\text{DMApy}]^+$ -based cations all have similar thermal conductivities. For the ILs containing the $[\text{emim}]^+$ cation, the significant difference among anions leads to a wider variation in measured thermal conductivity.

Table 3 summarizes experimentally measured densities for the four ILs containing the $[\text{Tf}_2\text{N}]^-$ anion for temperatures from 283 K to 363 K. Density is a linear function of temperature for these ILs within this temperature range. For the ILs based upon forms of the $[\text{DMApy}]^+$ cation, density decreases with increasing molecular weight.

Table 4 summarizes experimentally measured viscosities for $[\text{emim}][\text{DEP}]$, $[\text{emim}][\text{MeSO}_3]$, and $[\text{hDMApy}][\text{Tf}_2\text{N}]$ at several temperatures from 283 K to 343 K. Note that the measurements for $[\text{emim}][\text{MeSO}_3]$ below its melting point are for the subcooled liquid. Experimental viscosities were well-fit by the Vogel–Fulcher–Tamman (VFT) model,

$$\eta = \eta_0 \exp(k/(T - T_g)) \quad (3)$$

where η is the viscosity, T is the temperature, T_g is the glass transition temperature, and k and η_0 are fitted parameters. Table 5 lists newly measured values of T_g for these ILs and $[\text{emim}][\text{SCN}]$. Table 6 lists the remaining VFT parameters.

Table 7 summarizes experimentally measured heat capacities for $[\text{emim}][\text{DEP}]$, $[\text{hDMApy}][\text{Tf}_2\text{N}]$, and $[\text{hDMApy}][\text{Tf}_2\text{N}]$ at several temperatures from 298 K to 343 K. Within this temperature range, the heat capacity increases with increasing temperature for these ILs. The experimental heat capacity values for $[\text{hDMApy}][\text{Tf}_2\text{N}]$ reported here are slightly different from the values previously reported by our research group.⁶ This is because the purity of the commercial sample used in this study was higher than the purity of the $[\text{hDMApy}][\text{Tf}_2\text{N}]$ previously synthesized by our laboratory. The newly acquired data set is in better agreement with the computed heat capacity values and those reported by Diedrichs et al.²³

SIMULATION RESULTS

Results from simulations for each IL in this study were compared with the new experimental data described above and with results from the literature. To determine the effect of scaling the strength of electrostatic interactions, simulations for a given IL were generally performed using two different force field versions: the version with unscaled atomic partial charges (total ion charge of ± 1) and the version with 0.8-scaled charges (total ion charge of ± 0.8).

Figure 1 compares density versus temperature for both force field versions with experimental results for each IL in this study.^{9,24–26} Note that other published experimental density data for $[\text{emim}][\text{SCN}]$ are not shown for clarity,²⁷ but agree well with the density reported by Ficke et al. The 0.8-scaled force field consistently predicts lower densities than the unscaled force field. With the exception of $[\text{emim}][\text{TFA}]$, density predictions from the 0.8-scaled force field are in excellent agreement with experiment across the temperature range for which comparison was made, whereas the unscaled force fields generally predict densities that are higher than experimental densities by 2 % to 5 %. Compared to other properties, however, liquid densities are not especially sensitive to the quality of the force field. That is, different force fields may yield similar density predictions but give very different results for other properties. Therefore, a number of other properties were computed to judge the utility of these force fields.

Figure 2 compares thermal conductivities predicted by both force field versions (at 400 K) with experimental results (averaged over 300 to 375 K) for each IL in this study. The 0.8-scaled force fields consistently predict thermal conductivities slightly higher than experimental values and slightly lower than the values predicted by the unscaled force fields. Qualitative trends of thermal conductivity variation between different ILs are captured well by both force fields. As noted previously, experimental thermal conductivities for a given IL are essentially constant over the studied temperature range. Thermal conductivities predicted from simulations at 300 K were slightly higher than the values at 400 K, though we chose not to include the 300 K results because temperatures in the cold regions of the reverse nonequilibrium molecular dynamics (RNEMD) simulation boxes were comparable to experimental IL freezing points. See Supporting Information for an explanation of the RNEMD method. The 400 K simulation results were therefore deemed more reliable and are reported here.

Figure 3 compares heat capacity versus temperature predicted by both force field versions with experimental results for each IL in this study.^{6,23,24,28,29} For the $[\text{DMApy}]^+$ -based ILs (all with the $[\text{Tf}_2\text{N}]^-$ anion), heat capacity values predicted from simulations using force fields with unscaled charges were consistently lower than values from force fields with 0.8-scaled charges. For $[\text{emim}][\text{SCN}]$ and $[\text{emim}][\text{TFA}]$, force fields with unscaled charges predicted higher heat capacity values. For the remaining ILs, predictions from both force field versions were statistically equivalent. With the exception of $[\text{emim}][\text{TFA}]$, $[\text{hmim}][\text{Tf}_2\text{N}]$, and $[\text{hDMApy}][\text{Tf}_2\text{N}]$, predictions from both force field versions fall within experimental uncertainty. We also note that the differences between experimental heat capacity values obtained for the same ionic liquid but by different groups is often as large as the differences between the predictions of the two force fields and the differences between experimental and simulated heat capacity values. On the basis of this collection of simulation and experimental results, neither force field is clearly better at predicting heat capacity, but both yield reasonably accurate heat capacity values.

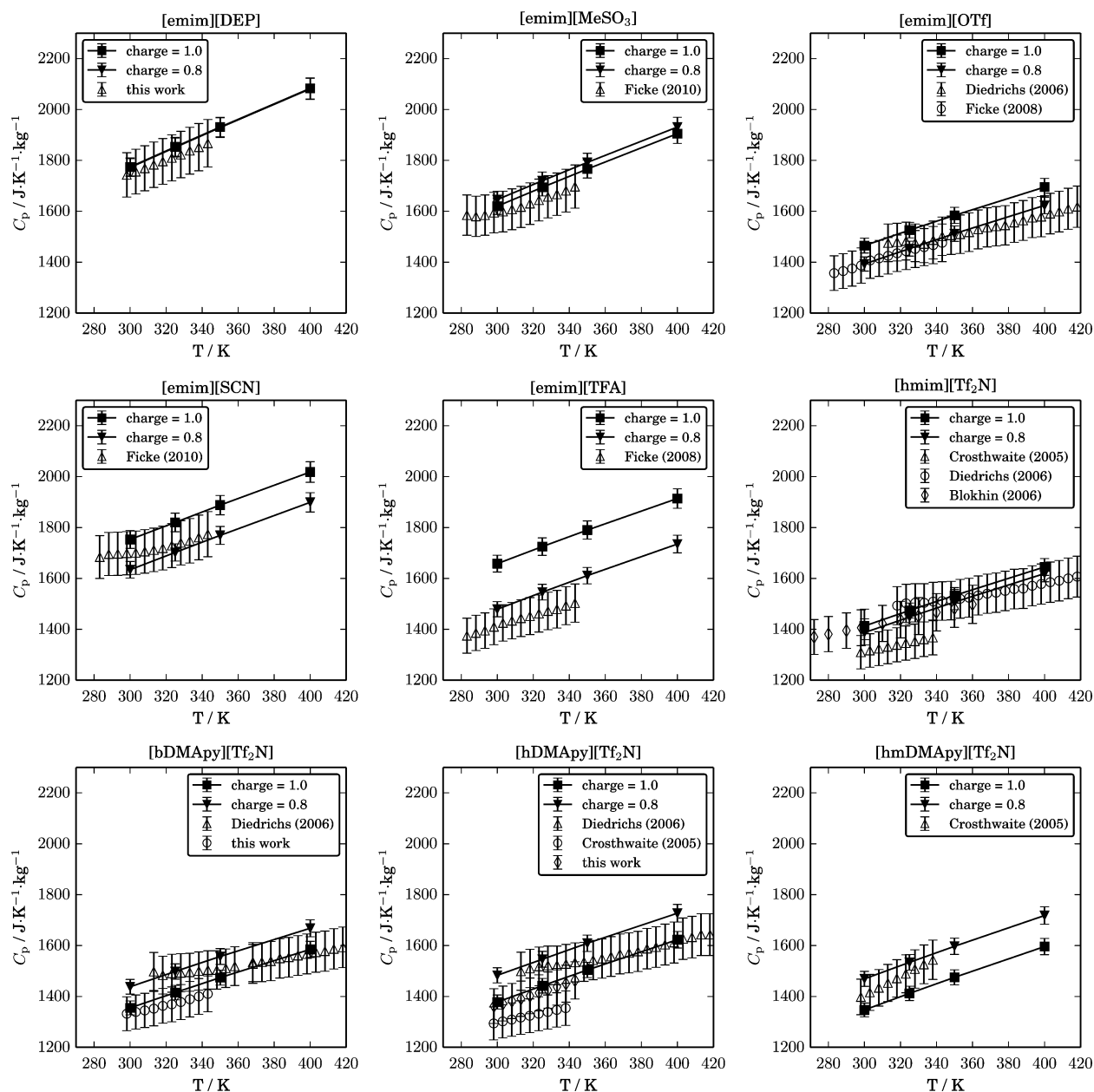


Figure 3. Computed heat capacity versus temperature (filled symbols) using force fields with unscaled and 0.8-scaled charges compared to experimental results (open symbols).^{6,23,24,28,29} Lines through simulation results are guides to the eye.

Figure 4 compares enthalpy of vaporization predicted by both force field versions with experimental results for [hmim][Tf₂N].^{30–32} Predictions from the force field with 0.8-scaled charges are slightly higher than most experimental data. The force field with unscaled charges clearly overestimates the enthalpy of vaporization by a large amount. On this basis, it appears that scaling charges by 0.8 results in more accurate, but still overly large, values for enthalpy of vaporization. This finding is consistent with recent results for other ionic liquids.^{33,34}

Figure 5 compares the Hildebrand solubility parameter predicted by both force field versions with experimental results for [hmim][Tf₂N].^{35,36} Consistent with expectations and the enthalpy of vaporization results described above, scaling the charges by 0.8 results in lower solubility parameter values relative to the unscaled case. Because of the limited amount of experimental data and the rather large discrepancy between experimental results at 300 K, it is difficult to draw conclusions

regarding the accuracy of the force fields. Both force field versions predict that the solubility parameter decreases slightly with increasing temperature, while the results of Marciniak show a slight increase in the solubility parameter with increasing temperature.³⁵ The two different sets of experimental results at 300 K bracket simulation results for both force fields; the data point from Lee is higher than both simulation results and the experimental result from Marciniak.^{35,36} From this it can be concluded that solubility parameters for [hmim][Tf₂N] have a fairly weak temperature dependence and range from 20 MPa^{1/2} to 25 MPa^{1/2}.

We also attempted to compute viscosities for the ILs via the Einstein method.³⁷ To do this, 30 ns NVT simulations at 400 K were carried out using force fields with 0.8-scaled charges for each IL in this study (results not shown). Previous experience had shown that viscosities calculated from force fields using unscaled charges were more than an order of magnitude larger

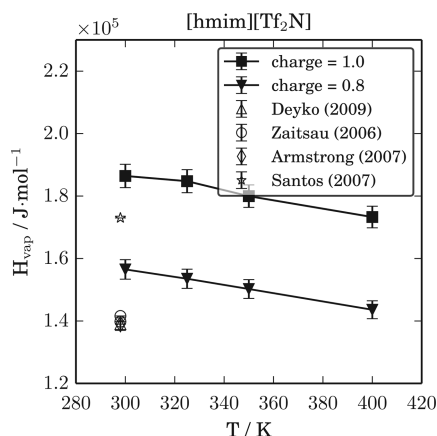


Figure 4. Enthalpy of vaporization versus temperature for [hmim]-[Tf₂N] from NPT molecular dynamics simulations (filled symbols) using force fields with unscaled and 0.8-scaled charges compared to experimental results (extrapolated to 300 K, open symbols).^{30–32} Lines are a guide to the eye.

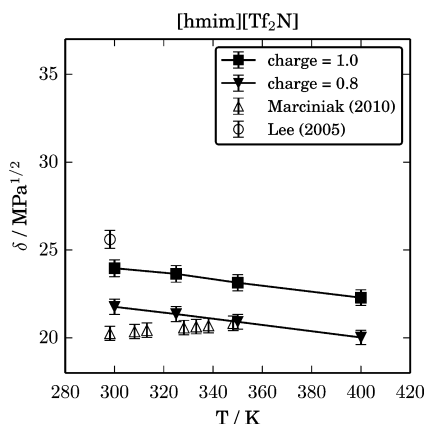


Figure 5. Hildebrand solubility parameter versus temperature for [hmim][Tf₂N] from NPT molecular dynamics simulations (filled symbols) using force fields with unscaled and 0.8-scaled charges compared to experimental results (open symbols).^{35,36} Lines through simulation results are guides to the eye.

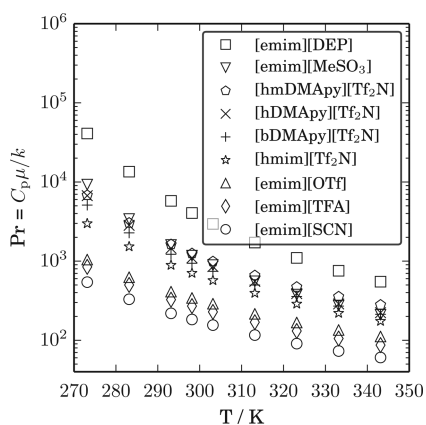


Figure 6. Prandtl number versus temperature calculated from experimental measurements of the present study. Ionic liquids in the legend are listed roughly in order of decreasing Prandtl number.

than experimental values, so no attempt was made to compute viscosities with the full-charge force fields. Even with 30 ns simulations, the standard uncertainty of the simulated viscosities

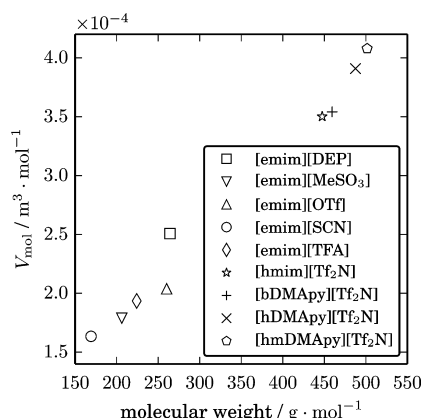


Figure 7. Molar volume at 400 K versus molecular weight from simulation results using force field with 0.8-scaled charges. Error bars are omitted for clarity.

approached the magnitude of the viscosity value, making quantitative comparison with experiment infeasible. In general, the simulated viscosities were ~5 times larger than experimental results.

On the basis of the comparisons with experiment described above, we conclude that the IL force fields with charges scaled by 0.8 generally provide more accurate results than the force fields with unscaled charges. The simulations do a quite good job capturing the thermodynamic properties of a wide range of ionic liquids. However, computed dynamical properties such as viscosity (and likely diffusivity) are still too slow relative to experimental results.

ANALYSIS

Figure 6 shows the experimental Prandtl number (the ratio of the heat capacity multiplied by the viscosity to the thermal conductivity) versus temperature for each IL in this study. Variations between the different ILs in this study span 2 orders of magnitude. Variations with temperature from 273 K to 343 K for a given IL span 1 order of magnitude. Nearly all of the variation in the Prandtl number is driven by variations in viscosity with temperature. Values of the Prandtl number range from 1000 or higher at the lowest temperatures down to order 100 at the highest temperatures. These values are typical of oils, and demonstrate that convective heat transfer is the dominant mechanism for ionic liquids.

Figure 7 shows molar volume at 400 K predicted by simulations using force fields with 0.8-scaled charges versus molecular weight for each IL in this study. For the ILs in this study, molar volume increases approximately linearly with increasing molecular weight.

Figure 8 shows heat capacity versus molar volume at 400 K predicted by simulations using force fields with 0.8-scaled charges for each IL in this study. For these ILs, heat capacity increases approximately linearly with increasing molar volume, consistent with previous experimental work.²⁴ More fundamentally, consideration of the relationship shown in Figure 8 shows that the amount of heat stored per unit volume is the same for every IL in this study.

Figure 9 shows thermal conductivity versus molar volume at 400 K predicted by simulations using force fields with 0.8-scaled charges for each IL in this study. Thermal conductivity generally decreases with increasing molar volume, and by extension, molecular weight. If bulk heat transfer within the IL is generally

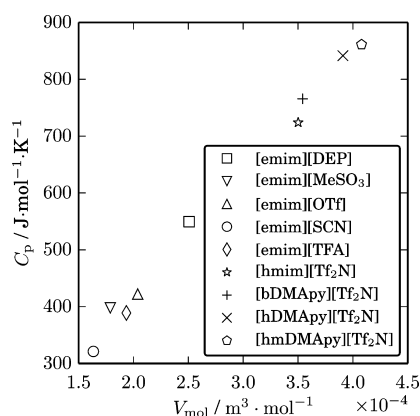


Figure 8. Heat capacity versus molar volume at 400 K from simulation results using force field with 0.8-scaled charges. Error bars are omitted for clarity.

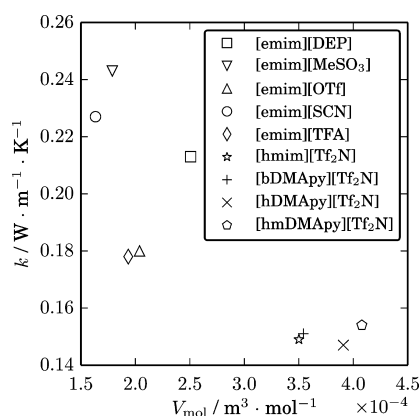


Figure 9. Thermal conductivity versus molar volume at 400 K from simulation results using force field with 0.8-scaled charges. Error bars are omitted for clarity.

limited by intermolecular collisions, this relationship is consistent with the fact that heavier molecules will generally have lower center-of-mass velocities at a given temperature, thereby extending the time between collisions and slowing heat transfer. The significant uncertainty in the trend displayed in Figure 9 (e.g., [emim][TFA] and [emim][OTf] compared with [emim][DEP]), suggests, however, that other IL characteristics also likely play a significant role in influencing thermal conductivity.

CONCLUSION

New experimental thermal conductivity, density, viscosity, glass transition temperature, and heat capacity values were measured for nine ionic liquids (ILs): [emim][TFA], [emim][OTf], [emim][DEP], [emim][MeSO₃], [emim][SCN], [hmim][Tf₂N], [bDMApy][Tf₂N], [hDMApy][Tf₂N], and [hmDMApy][Tf₂N].

Classical molecular mechanics force fields were developed for the necessary cations and anions and used to calculate thermodynamic and transport properties for these ILs using molecular dynamics. Two versions of each force field were developed: the first with conventional integer charges of ± 1 on cations/anions, and the second with all partial charges scaled by 0.8.

The force fields with total charges of ± 0.8 generally gave better agreement with experimental measurements. Very good agreement was attained for density and heat capacity. Simulated

values for thermal conductivity slightly overpredicted experimental results but captured trends between different ILs very well. Viscosities calculated from simulation were much better for the 0.8-scaled force fields, but still significantly overpredicted experimental values. Experimental values for enthalpy of vaporization and the Hildebrand solubility parameter also generally show better agreement with the 0.8-scaled force fields.

Within this set of ILs, molar heat capacity was proportional to molar volume, varying in magnitude by a factor of 3. Thermal conductivity showed a weaker and generally decreasing trend with increasing molar volume, varying by a factor of 0.6. In contrast, measured viscosities varied by 2 orders of magnitude between different ILs. The Prandtl number showed wide variation among different ILs and a strong temperature dependence, driven mainly by strong dependence of viscosity on temperature. For all the ILs, the Prandtl number is large, suggesting that convective effects will dominate in heat transfer applications.

ASSOCIATED CONTENT

Supporting Information

Experimental water content, descriptions of the simulation methods, simulation force field parameters, and example input scripts. This material is available free of charge via the Internet at <http://pubs.acs.org>.

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Notes

The authors declare no competing financial interest.

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